

Unit 12, Lesson 6: Solving Multi-Step Problems Involving Discounts, Taxes, Tips, or Fees



Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.



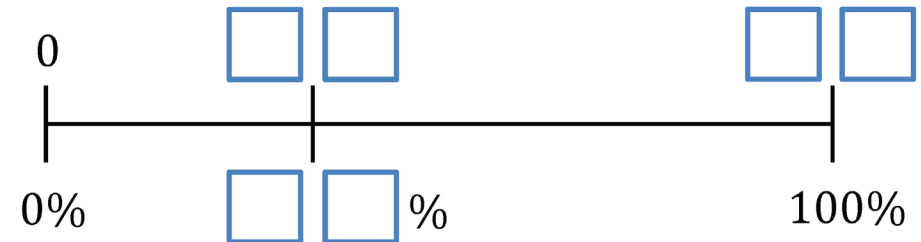
Learning Target

- ☐ I can solve multi-step problems that involve discounts, markups, taxes, tips, or fees.



12.6.1 Warm-Up: Percentages on a Linear Model

- Fill in the blue boxes using the digits 1 – 9, one time each at most, to create an accurate number line.



- There are multiple possible solutions.
 - Find at least one other solution.
 - Explain whether using 47% for the percentage makes sense.



12.6.2 Exploration: Finding Sales Tax Rates and Total Amounts

1. Cities often have different sales tax rates based on local government decisions.
 - a. Discuss with your partner how you might find the percentage of sales tax if you are given the price of an item and the amount of sales tax paid for the item. Summarize your discussion.
 - b. With your partner, complete the two tables to determine the sales tax rates for City A and City B.

City A			
Item	Price (\$)	Sales Tax (\$)	Sales Tax Rate (%)
Paper Towels	8.99	0.63	
Laundry Soap	11.83	0.83	

City B			
Item	Price (\$)	Sales Tax (\$)	Sales Tax Rate (%)
Paper Towels	8.99	0.54	
Laundry Soap	11.83	0.71	

- c. Determine the total cost, with tax, of paper towels and laundry soap in City A.
- d. The pre-tax cost of 5 beach towels in City A is \$54.95. Determine the total cost, with tax, of 5 beach towels in City A.
- e. If the subtotal for the purchase of a new bathing suit in City B is \$47.97, what is the total cost of the bathing suit with tax?
- f. Find the total cost, with tax, on paper towels, laundry soap, and a new bathing suit in City B.

2. Jada and Corrin had lunch at a local restaurant. The final bill for their meal is shown.

They decide to tip their server 18% of the subtotal. Jada says the tip should be \$7.56, but Corrin disagrees. Corrin calculated the tip to be \$8.28.

- a. Explain who calculated the correct tip.

Date: Sep. 12th
Time: 6:55 PM
Server: # 27

Bread Stix	9.50
Chicken Parm	15.50
Chef Salad	12.00
Lemon Soda	2.00
Tea	3.00

Subtotal	42.00
Sales Tax	3.99
Total	45.99

- b. What is the total amount, including the tip, that the friends paid at the restaurant?



12.6.3 Collaboration: Taxes, Tips, and Totals

Work through each situation with your partner. Be sure to show your work.

1. Xavier visits an Italian restaurant for dinner.

- a. If Xavier tips 20% on the subtotal \$24.95, how much tip will he pay?

Date: Sep 19th
Time: 6:04 PM
Server: Flo

Bread Stix	9.50
Ravioli Bites	10.50
Cheesecake	4.95

Subtotal	24.95
Sales Tax	_____
Tip	_____
Total	_____

- b. The city has a sales tax rate of 9.5%. How much in tax will the customer pay on the subtotal of \$24.95?
- c. What is the total Xavier spent on his meal, including tip and tax?

2. A sporting goods store purchased baseball hats from a supplier for \$24 each and marked them up to \$35.

a. Calculate the percentage of the markup.

b. A customer orders 4 baseball hats from the store online. What is the subtotal of the 4 hats?

c. If the sales tax is 6.25% of the subtotal and there is a \$7.00 shipping fee applied after tax, what is the total amount the customer paid?

3. Leilani found a 30% off discount code for an NXT True Hybrid Surfboard. The original cost of the surfboard is \$700. If the company does not charge tax but instead charges a \$150 shipping fee, how much does the surfboard cost in total?



12.6.4 Your Turn!

1. A car dealership pays a wholesale price of \$12,000 to purchase a vehicle. The car dealership wants to make a 32% profit.



- How much money is added to the wholesale price to make a 32% profit?
- After the markup, what is the total retail price of the vehicle?
- The sales tax in the city it is sold is 7.5%. After tax, what will the customer pay for the car?

2. The table shows the pre-tax cost of items at a convenience store where the sales tax rate is 4.6%.

Item	Cost (\$)
Milk	3.75
Bag of Ice	2.50
Aspirin – 24 tablets	8.25
Snack Mix	3.75
Candy Bar	1.25
Batteries	6.75

- Cal stopped at the convenience store and picked up a bag of ice and batteries. What was his total cost, with tax?
- Ms. Isales has \$10.00 in her wallet. Does she have enough to buy aspirin and a candy bar? Explain your reasoning using mathematics.

3. A local restaurant automatically adds a 21% gratuity (tip), pre-tax, to each bill for parties over 10. The table shows the bills due for several parties over 10. Complete the table.

Party	Subtotal (\$)	Sales Tax (\$)	Gratuity (\$)	Total Amount of Bill
Flores Party	93.75	5.86		
Atkinson Party	176.99		37.17	
Mikelson Party	135.50			172.43



12.6.5 Wrap-Up: Ticket Out the Door

1. Diesel has been waiting patiently for the Evolve Carbon All Terrain Skateboard to go on sale. For his birthday, he received a 25% off discount. The original price of the skateboard is \$1,999.99 and the company has a 5% sales tax, along with a flat fee of \$35 for shipping.

What is the total amount Diesel needs to purchase his dream skateboard?